1. Objective

The objective of this lab session was to install and set up the C++ programming environment in Visual Studio Code (VS Code) on Microsoft Windows. This included installing the MinGW compiler, configuring environment variables, and adding necessary extensions to run C++ programs efficiently.

2. Tools and Software Used

- Visual Studio Code (IDE)
- MinGW-w64 Compiler (for C++ compilation)
- C/C++ Extension by Microsoft
- Code Runner Extension by Jun Han

3. Procedure

Step 1: Install Visual Studio Code

- Downloaded VS Code from https://code.visualstudio.com/Download.

- Installed it by following the on-screen instructions.

Step 2: Install MinGW Compiler

- Downloaded MinGW-w64 from Source forge.
- Installed the compiler and selected the package.

- Applied changes to finalize installation.

Step 3: Configure Environment Variables

- Located the compiler path (e.g.,).
- Added this path to System Environment Variables → Path.
- Saved changes to ensure could be accessed globally.

Step 4: Install Extensions in VS Code

- Installed C/C++ Extension by Microsoft.
- Installed Code Runner Extension by Jun Han.
- Restarted VS Code to activate extensions.

4. Lab Task

The lab task required:

1. Forming a group of three students.
2. Completing installation and setup of C++ in VS Code.
3. Running a simple program to display group members' names.

EXAMPLE PROGRAM:

#include <iostream>

```
using namespace std;
int main() {
cout << "Group Members:" << endl;
  cout << "1st " << endl;
   cout << "2. [Second Member]" << endl;
   cout << "3. [Third Member]" << endl;
   return 0;
}
```

5. OUTPUT

Group Members:

1.{first member}

2. [Second Member]

3. [Third Member]

6. Observations

- The installation process was straightforward but required careful attention to environment variable configuration.
- Without adding the MinGW path to system variables, VS Code could not compile C++ programs.
- Extensions like Code Runner simplified execution by allowing direct running of code within VS Code.

7. Conclusion

The lab successfully achieved its objective. C++ was installed and configured in VS Code, and a sample program was executed to display group members' names. This setup will be used for future programming labs a**nd assignments.**